

Input data

Household attributes

Household size
Number of workers
Number of vehicles
Number of bikes
Number of e-bikes
Number of motorcycles
Number of EMVs
Household annual income
Presence of students

Individual attributes

Household size
Number of workers
Number of vehicles
Number of bikes
Number of e-bikes
Number of motorcycles

1. Household segmentation

(To handle “one-to-many” problem in hierarchical data)

Extended household attribute

HDBSCAN clustering

KNN-based noise point assignment

Clustering label & profile

2. Hierarchical Diffusion Transformer Model (HiDiT)

Up-level household diffusion transformer

(To generate household attributes from noise)

Household noise

Diffusion transformer

Household attributes

Graph-related module

(To explicitly model intra-household relationships)



Household attributes

GraphVAE & HGT

Household structure representation

Down-level individual diffusion transformer

(To generate individual attributes from noise)

$\oplus$

Individual noise

Diffusion transformer

Individual attributes

3. Energy-based guidance

(To ensure multi-scale marginal alignment)

Known Multi-scale marginal constraints

Energy-based guidance function

Multi-scale gradient calculation

Conflict resolution between gradient

Sampling process

4. Regularization strategies

(To control sampling & structural zeros)

Generated population

Bayesian network

VAE

Valid flag

Synthetic population

Household attributes

Household size
Number of workers
Number of vehicles
Number of bikes
Number of e-bikes
Number of motorcycles
Number of EMVs
Household annual income
Presence of students

Individual attributes

Household size
Number of workers
Number of vehicles
Number of bikes
Number of e-bikes
Number of motorcycles